SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10	**Course Title:**	Software Engineering
CO Assessed:	CO5	**Assessment:**	Assessment Tool 1 – **Research Paper Review / Literature Survey**
Weightage:	20%	**Total Marks:**	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:		**Reg. No.:**	
Date:		**Duration:**	60 minutes

Code of Conduct

I __________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __________________________

Annexure A – Research Paper Review / Literature Survey

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **conduct a literature survey and review relevant research papers** to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem.

The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statemen tfrom reputed journals, conferences, or scholarly databases.

2.Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.

3.Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.

4.Identify the **limitations and challenges** of the existing research approaches.

5.Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.

6.Discuss the **possible improvements or future research directions** based on the identified research gap.

7.Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.

8.Provide proper **citations and references** using a consistent referencing style.

Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions	Proposes practical	Proposes relevant	Provides	Future

& Relevance(15%)	and innovative future directions directly relevant to the assigned problem statement	future directions	general future suggestions	directions are missing or unrelated
Documentation, Citations & References(15%)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	**Mark**
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5
Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25

1. INTRODUCTION

Modern warehouses are increasingly adopting advanced technologies such as Artificial Intelligence (AI), Internet of Things (IoT), Radio Frequency Identification (RFID), cloud computing, and Autonomous Mobile Robots (AMRs) to improve inventory accuracy, storage utilization, order fulfillment, and overall operational efficiency.

Traditional warehouse operations often depend on manual inventory counting, barcode scanning, fixed storage arrangements, and human-operated material handling. These methods can result in inventory discrepancies, misplaced products, inefficient utilization of storage space, delayed order fulfillment, and limited real-time visibility of inventory movement.

RFID technology enables automatic identification and tracking of products, while IoT devices provide continuous collection of warehouse information. AI algorithms can analyze this information and support intelligent decisions such as storage allocation, inventory prediction, order prioritization, order batching, and picking optimization. AMRs can further automate the movement of materials between storage, picking, packing, and dispatch areas.

Therefore, an intelligent warehouse automation platform integrating RFID, IoT, AI, cloud computing, and robotic systems can provide a more efficient and data-driven approach to warehouse management.

2. PROBLEM STATEMENT

Manual warehouse operations often lead to inventory errors, inefficient storage utilization, delayed order fulfillment, and increased operational costs. Existing warehouse management systems may provide inventory records but have limited ability to continuously monitor product movement and make intelligent operational decisions.

The proposed intelligent warehouse platform should:

1. Track inventory using RFID.
2. Collect warehouse information using IoT.
3. Monitor inventory movement in real time.
4. Optimize storage locations using AI.
5. Identify inventory discrepancies.
6. Improve order-picking efficiency.
7. Automate inventory updates.
8. Support autonomous material movement.
9. Provide real-time warehouse monitoring.
10. Generate warehouse performance reports.

The overall objective is to create an intelligent and connected warehouse environment that improves productivity while reducing operational costs.

3. AIM OF THE LITERATURE SURVEY

The aim of this literature survey is to study existing research related to AI-based warehouse automation and inventory management.

The review focuses on:

1. RFID-based inventory tracking.
2. IoT-based warehouse monitoring.
3. AI-based storage optimization.
4. AI-based order picking.
5. Deep Reinforcement Learning in warehouses.
6. Autonomous Mobile Robots.
7. Real-time inventory management.
8. Cloud-based warehouse systems.
9. Intelligent warehouse decision-making.
10. Integration of Industry 4.0 technologies.

The literature review is used to understand existing methodologies, technologies, datasets, findings, limitations, and research gaps.

4. OBJECTIVES

The major objectives of the proposed warehouse automation platform are:

1. To automate warehouse inventory tracking.
2. To optimize storage allocation using AI.
3. To monitor inventory movement in real time.
4. To improve order fulfillment efficiency.
5. To reduce inventory discrepancies.
6. To generate warehouse performance reports.
7. To integrate RFID-based inventory management.
8. To use IoT for continuous warehouse monitoring.
9. To support automated material movement.
10. To improve overall warehouse productivity.

5. LITERATURE SURVEY

RESEARCH PAPER 1

RFID in the Warehouse: A Literature Analysis (1995–2010) of Its Applications, Benefits, Challenges and Future Trends

Authors: Ming K. Lim, Witold Bahr and Stephen C. H. Leung

Year: 2013

Journal: International Journal of Production Economics

Research Problem

The research investigates the application of RFID technology in warehouse operations and studies its benefits, challenges, applications, and future research directions.

Methodology

The authors conducted a comprehensive literature analysis of research published between 1995 and 2010. The studies were analyzed according to RFID applications, warehouse operations, benefits, and implementation challenges.

Technologies

- RFID tags
- RFID readers
- RFID middleware
- Warehouse information systems

Major Findings

RFID can improve inventory visibility, product identification, inventory traceability, automated stock counting, receiving operations, inventory accuracy, and warehouse information collection.

RFID reduces the dependency on manual product identification and can provide more accurate information about inventory movement.

Limitations

The research primarily focuses on RFID and does not provide a complete integrated AI, IoT, and autonomous robot architecture.

Research Gap

RFID can be integrated with AI and IoT to transform automatically collected inventory information into intelligent storage and order-picking decisions.

6. RESEARCH PAPER 2

A Systematic Literature Review of RFID in Supply Chain Management

Year: 2021

Journal: Journal of Enterprise Information Management

Research Problem

The research investigates the development and application of RFID technology within supply-chain management.

Methodology

The researchers conducted a systematic literature review and bibliometric analysis of research articles.

The analysis considered:

- Research trends
- Keywords
- Research themes
- RFID applications
- Relationships with emerging technologies

Technologies

- RFID
- IoT
- Machine Learning
- Supply-chain information systems
- Data analytics

Major Findings

The research indicates that RFID is increasingly being connected with technologies such as IoT and machine learning. RFID can provide the data required for improved supply-chain visibility and decision-making.

Limitations

The study focuses on supply-chain management as a broader area rather than developing a complete warehouse automation platform.

Research Gap

There is a need for integrated systems that connect RFID-generated inventory information with AI-based decision-making and warehouse automation.

7. RESEARCH PAPER 3

Applications and Methodologies of Internet of Things in Warehouses and Inventory Management: A Systematic Literature Review

Year: 2025

Journal: Procedia Computer Science

Research Problem

The study investigates the use of IoT technologies in warehouse and inventory management.

Methodology

A systematic literature review was performed on research related to IoT applications in warehouses and inventory management.

The study examined:

- IoT architectures
- Warehouse applications
- Inventory monitoring
- Technologies
- Models
- Research trends

Technologies

- IoT
- Sensors
- RFID
- Communication networks
- Cloud platforms

Major Findings

IoT can provide continuous data collection and real-time visibility into warehouse operations. IoT technologies can support inventory monitoring, warehouse decision-making, and connected warehouse equipment.

Limitations

Many studies focus on theoretical frameworks and algorithms, while practical real-world implementation is comparatively limited.

Research Gap

A practical prototype integrating IoT with RFID, AI, and a real-time warehouse dashboard can address this implementation gap.

8. RESEARCH PAPER 4

Deep Reinforcement Learning for Dynamic Order Picking in Warehouse Operations

Authors: Sasan Mahmoudinazlou, Abhay Sobhanan, Hadi Charkhgard, Ali Eshragh and George Dunn

Year: 2025

Journal: Computers & Operations Research

Research Problem

Warehouse order-picking operations become difficult when orders arrive continuously and warehouse conditions change dynamically.

Traditional fixed picking and routing methods may not respond effectively to changing workloads.

Methodology

The research uses:

- Markov Decision Process
- Deep Reinforcement Learning
- Dynamic order information
- Autonomous picking devices
- Route optimization

Major Findings

The proposed Deep Reinforcement Learning approach dynamically optimizes order-picking decisions. The reported experiments demonstrated improved order fulfillment compared with benchmark approaches under the studied conditions.

Limitations

The research focuses mainly on dynamic order picking and does not implement a complete RFID-based inventory management system.

Research Gap

Inventory tracking, storage allocation, order picking, and autonomous robot operations can be integrated into one intelligent warehouse platform.

9. RESEARCH PAPER 5

Solving the Online Batching Problem Using Deep Reinforcement Learning

Year: 2021

Journal: Computers & Industrial Engineering

Research Problem

Warehouse orders arrive continuously, making it difficult to determine which orders should be grouped together for efficient picking.

Methodology

The research uses Deep Reinforcement Learning to determine suitable order-batching decisions.

The system learns:

- When orders should be processed.
- Which orders should be combined.
- How orders can be grouped efficiently.

Major Findings

The approach can improve order-batching decisions and help reduce delays associated with warehouse order processing.

Limitations

The research focuses mainly on order batching rather than complete inventory tracking and warehouse automation.

Research Gap

Order batching can be connected with RFID-based inventory information and real-time warehouse data to create a more complete decision-making system.

10. RESEARCH PAPER 6

Deep Reinforcement Learning for Two-Sided Online Bipartite Matching in Collaborative Order Picking

Authors: Luca Begnardi, Hendrik Baier, Willem van Jaarsveld and Yingqian Zhang

Year: 2024

Conference: Asian Conference on Machine Learning

Research Problem

Modern warehouses may use both human workers and robots. Efficiently assigning picking tasks between humans and robots is challenging.

Methodology

The research investigates:

- Deep Reinforcement Learning
- Graph Neural Networks
- Simulation
- Human-robot coordination
- Order-picking task assignment

Major Findings

The proposed methods were reported to outperform selected baseline approaches in the studied scenarios.

Limitations

The research focuses on collaborative order picking rather than complete RFID-based inventory management.

Research Gap

RFID inventory information could be integrated with AI task assignment so that robot and worker tasks are based on real-time inventory conditions.

11. RESEARCH PAPER 7

Dynamic Multi-Tour Order Picking in an Automotive-Part Warehouse Based on Attention-Aware Deep Reinforcement Learning

Year: 2025

Journal: Robotics and Computer-Integrated Manufacturing

Research Problem

Dynamic warehouse environments require picking routes to adapt to changing orders and warehouse conditions.

Methodology

The research uses:

- Deep Reinforcement Learning
- Attention mechanisms
- Markov Decision Process
- Proximal Policy Optimization
- Dynamic warehouse information

Major Findings

The proposed approach improves dynamic order-picking decisions compared with selected alternative approaches in the reported experiments.

Limitations

The study concentrates on dynamic order picking and does not provide a complete RFID-IoT inventory management platform.

Research Gap

Real-time inventory information from RFID and IoT devices can be integrated into AI-based dynamic picking decisions.

12. COMPARATIVE LITERATURE SURVEY

Author / Year	**Research Problem**	**Methodology**	**Technologies**	**Key Findings**	**Limitations**	**Research Gap**
Lim et al., 2013	RFID applications in warehouses	Literature analysis	RFID	Improves inventory visibility and traceability	RFID-focused	AI + IoT + robotics integration
al., 2013	RFID applications in warehouses	Literature analysis	RFID	Improves inventory visibility and tracLim eteability	RFID-focused	AI + IoT + robotics integration
2021 RFID Review	RFID in supply chains	Systematic literature review	RFID, IoT, ML	RFID increasingly connects with IoT and ML	Broad supply-chain focus	Complete intelligent warehouse
2025 IoT Review	IoT in	Systematic	IoT, RFID,	Improves real-	Limited	Practical

	warehouses	literature review	Cloud	time warehouse monitoring	practical implementations	integrated prototype
Mahmoudinazlou et al., 2025	Dynamic order picking	Deep Reinforcement Learning	DRL, autonomous picking	Improves dynamic order fulfillment	Picking-focused	Integrate inventory tracking
2021 Online Batching	Order batching	Deep Reinforcement Learning	DRL	Improves batching decisions	Batching-focused	Combine with RFID and IoT
Begnardi et al., 2024	Human-robot picking	DRL + GNN	Robotics, DRL	Improves task coordination	Picking-focused	Integrate inventory information

13. COMPARATIVE ANALYSIS

RFID-Based Systems

RFID is mainly responsible for automatically identifying and tracking warehouse items.

Advantages

- Automatic identification.
- Reduced manual scanning.
- Real-time product visibility.
- Improved traceability.
- Faster inventory counting.
- Reduced human errors.

Limitations

RFID mainly answers:

"What item is present and where was it detected?"

It does not independently determine:

- Where the item should be stored.
- Which order should be prioritized.
- Which robot should pick it.
- Which route is optimal.

Therefore, AI is required for intelligent decision-making.

14. IoT-BASED SYSTEMS

IoT connects warehouse devices and continuously collects operational information.

Examples

- RFID readers
- Temperature sensors
- Weight sensors
- Location sensors
- Robot status sensors
- Warehouse gateways

Advantages

- Real-time monitoring.
- Continuous data collection.
- Remote monitoring.
- Cloud connectivity.
- Better operational visibility.

Limitation

IoT primarily collects data. AI or analytics is required to convert the collected data into intelligent decisions.

15. AI-BASED SYSTEMS

AI can analyze warehouse information and recommend appropriate decisions.

AI can be used for:

1. Storage allocation.
2. Demand prediction.
3. Inventory classification.
4. Order batching.
5. Picking optimization.
6. Robot task assignment.
7. Route optimization.
8. Inventory anomaly detection.
9. Low-stock prediction.
10. Warehouse performance analysis.

16. AUTONOMOUS MOBILE ROBOTS

Autonomous Mobile Robots can automate the physical movement of products.

A typical operation is:

Storage Area → AMR → Picking Area → Packing Area → Dispatch Area

AMRs can improve:

- Material movement.
- Picking efficiency.
- Worker productivity.
- Warehouse flexibility.
- Order fulfillment.
- Operational efficiency.

For a student project, a physical AMR is not necessary. The robot can be represented through a **Python-based warehouse grid simulation**.

17. IDENTIFIED RESEARCH GAPS

Research Gap 1 – Lack of Complete Integration

Many studies focus on individual technologies such as RFID, IoT, AI, or robotics rather than integrating them into one complete platform.

Research Gap 2 – Limited Real-Time AI Decision-Making

RFID and IoT can collect real-time information, but this information is not always directly connected to an AI decision-making system.

Research Gap 3 – Separate Optimization

Existing studies frequently focus on one specific warehouse operation such as:

- Picking
- Routing
- Batching
- Storage allocation

A complete intelligent platform should coordinate multiple operations.

Research Gap 4 – Limited Practical Prototypes

Many studies are based on theoretical models, simulations, or algorithms. There is scope for developing affordable practical prototypes suitable for smaller warehouses.

Research Gap 5 – Inventory-to-Robot Integration

A direct connection between inventory tracking and robotic task assignment can improve warehouse automation.

18. PROPOSED SYSTEM

Intelligent AI-Based Warehouse Automation and Inventory Tracking Platform

The proposed system integrates:

RFID + IoT + AI + Cloud Database + Dashboard + AMR Simulation

19. PROPOSED SYSTEM ARCHITECTURE

```
                    WAREHOUSE
                        |
             -----------------------
             |                     |
         RFID Tags            IoT Sensors
             |                     |
             -----------+-----------
                        |
                 RFID / IoT Gateway
                        |
                        v
                 Cloud / Database
                        |
                        v
                 AI Decision Engine
                        |
          --------------+--------------
          |             |             |
          v             v             v
      Storage       Inventory       Order
      Allocation    Prediction      Priority
          |             |             |
          --------------+--------------
                        |
                        v
                   AMR Module
                        |
                        v
              Warehouse Movement
                        |
                        v
               Real-Time Dashboard
```

20. PROPOSED METHODOLOGY

Step 1 – Product Registration

Each product is assigned a unique RFID tag.

Product ID	Product	Quantity	RFID Tag
P001	Laptop	20	RFID001
P002	Keyboard	50	RFID002
P003	Mouse	75	RFID003
P004	Monitor	15	RFID004

Step 2 – RFID-Based Inventory Detection

RFID readers detect products entering or leaving different warehouse zones.

The system records:

- Product ID
- Location
- Date
- Time
- Entry/exit status
- Quantity

This information is automatically transferred to the database.

21. STEP 3 – DATABASE UPDATE

Product	Previous Quantity	Movement	New Quantity
Laptop	20	-2	18
Keyboard	50	+10	60
Mouse	75	-5	70

The database automatically updates the inventory instead of requiring manual entry.

22. STEP 4 – AI-BASED STORAGE ALLOCATION

The AI system analyzes:

- Product demand.
- Current inventory.
- Product size.
- Storage availability.
- Order frequency.

- Picking frequency.
- Distance from dispatch.

Product	Demand Level	Recommended Storage
Laptop	High	Near picking area
Keyboard	High	Near picking area
Monitor	Medium	Middle zone
Seasonal Product	Low	Far storage zone

High-demand products can be placed closer to the picking and dispatch areas.

23. STEP 5 – ORDER PROCESSING

Suppose the warehouse receives:

Product	Required Quantity
Laptop	2
Mouse	3
Keyboard	1

The system checks the database and verifies whether the required quantity is available.

24. STEP 6 – AI-BASED PICKING DECISION

The AI system determines:

1. Required products.
2. Product storage locations.
3. Picking priority.
4. Efficient route.
5. Robot assignment.

25. STEP 7 – AMR SIMULATION

The AMR can be simulated using a warehouse grid.

```
START
   ↓
Laptop Storage
   ↓
Mouse Storage
   ↓
Keyboard Storage
```

```
   ↓
Packing Area
   ↓
Dispatch
```

The simulation can display the robot moving from one warehouse location to another.

27. STEP 9 – LOW-STOCK DETECTION

The system automatically generates an alert when stock falls below a predefined threshold.

```
LOW STOCK ALERT

Product: Laptop
Current Stock: 4
Reorder Level: 5
Status: REORDER REQUIRED
```

28. STEP 10 – INVENTORY ANOMALY DETECTION

The system can compare expected inventory with RFID-detected inventory.

```
Expected Quantity = 50
RFID Detected Quantity = 43

Difference = 7

Status = INVENTORY DISCREPANCY
```

This can help identify:

- Missing products.
- Incorrect records.
- Unauthorized movement.
- Data-entry errors.

29. AI IMPLEMENTATION

A simple and practical AI implementation can be used instead of complicated algorithms.

AI Model 1 – Demand Prediction

The model predicts product demand using historical data.

Possible algorithms:

- Random Forest

- Decision Tree
- XGBoost

Input features can include:

- Previous sales.
- Current stock.
- Weekly demand.
- Monthly demand.
- Product category.

Output:

High / Medium / Low Demand

AI Model 2 – Storage Recommendation

The system recommends the best storage zone based on:

- Demand frequency.
- Product size.
- Current stock.
- Picking frequency.
- Distance from dispatch.

AI Model 3 – Inventory Anomaly Detection

The system detects unusual differences between expected and observed stock.

Example:

Expected = 100

Detected = 87

Difference = 13

The system generates an inventory discrepancy alert.

30. TECHNOLOGIES AND TOOLS

Component	Suggested Technology
RFID	RC522 / UHF RFID
Microcontroller	ESP32
IoT Communication	Wi-Fi / MQTT
Database	MySQL / Firebase
Programming	Python
AI	Scikit-learn
Data Processing	Pandas
Dashboard	Streamlit / Power BI
Visualization	Matplotlib / Plotly
AMR	Python Simulation
Development	VS Code / Google Colab

31. PERFORMANCE METRICS

The proposed system can be evaluated using the following metrics.

Inventory Accuracy

Inventory Accuracy = Correct Inventory Records / Total Inventory Records × 100

Order Fulfillment Rate

Order Fulfillment Rate = Completed Orders / Total Orders × 100

Average Picking Time

Average Picking Time = Total Picking Time / Number of Orders

Warehouse Utilization

Warehouse Utilization = Used Storage Capacity / Total Storage Capacity × 100

Inventory Discrepancy Rate

Inventory Discrepancy Rate = Discrepant Items / Total Items × 100

These metrics can be used to compare the proposed system with a conventional manual warehouse process.

32. EXPECTED OUTCOMES

The proposed system is expected to achieve:

1. Improved inventory accuracy.

2. Automated inventory tracking.
3. Real-time inventory visibility.
4. Reduced manual data entry.
5. Improved storage utilization.
6. Faster order processing.
7. Reduced picking distance.
8. Better warehouse productivity.
9. Automatic low-stock alerts.
10. Inventory discrepancy detection.
11. Real-time warehouse monitoring.
12. Data-driven warehouse management.
13. Improved logistics efficiency.
14. Reduced operational costs.
15. Improved customer satisfaction.

33. FUTURE SCOPE

The proposed system can be further enhanced by implementing:

1. Computer vision for product recognition.
2. Multiple AMRs.
3. Digital twin technology.
4. Predictive maintenance.
5. Advanced demand forecasting.
6. Reinforcement learning.
7. Edge AI.
8. Multi-warehouse optimization.
9. Blockchain-based product traceability.
10. Voice-controlled warehouse systems.
11. Real-time 3D warehouse visualization.

34. OVERALL RESEARCH GAP STATEMENT

Existing research demonstrates the individual benefits of RFID-based inventory tracking, IoT-based warehouse monitoring, AI-based optimization, and robotic order picking. However, these technologies are frequently studied as separate components. There is an opportunity to develop a practical integrated warehouse platform that connects real-time RFID and IoT inventory information with AI-based storage allocation, inventory prediction, order optimization, and autonomous robot task execution.

The proposed system addresses this gap by integrating **RFID, IoT, AI, cloud/database technology, real-time dashboards, and AMR simulation** into a unified warehouse automation framework.

35. CONCLUSION

The literature survey demonstrates that Industry 4.0 technologies can significantly improve warehouse operations. RFID enables automatic product identification and inventory tracking, IoT provides continuous monitoring and communication, AI supports intelligent decision-making, and autonomous mobile robots can automate material transportation.

However, the reviewed research indicates that these technologies are often investigated separately, creating an opportunity for an integrated solution. The proposed **Intelligent AI-Based Warehouse Automation and Inventory Tracking Platform** combines these technologies to provide real-time inventory visibility, intelligent storage allocation, automated inventory updates, efficient order processing, inventory discrepancy detection, and simulated autonomous material movement.

The proposed system can therefore improve warehouse accuracy, efficiency, productivity, and operational decision-making while providing a practical and achievable prototype for academic implementation.

36. REFERENCES

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